

Blurred Lens

Does an AI image model paint poorer countries in warmer colour? 61,488 generated photographs say the honest answer is more interesting than yes.

Haydn Stucker · AIPI 590: Explainable AI · Duke University

61,488
images generated

60 × 6
countries × kinds of place

53 of 60
warmer than the model's own default

0
failures or refusals

1 The problem

Film colours places. A sepia wash over Mexico; dust and amber for “somewhere dangerous.” A yellow filter tells an audience a place is poor, hot and unsafe before a word is spoken.

Image models learn from pictures like those. If one inherited the habit, every picture it makes of a country carries a quiet judgement — and no single image can show you.

2 How we looked

1 Ask everyone the same thing

One sentence, one fixed camera, 84 times per prompt.

2 Take away the easy excuse

Run it again pinned to noon under a clear sky. The pin names no colour.

3 Give it a zero

The same places with no country named — the model's own default.

4 Pick countries that disagree

15 per income group, so income and latitude pull apart ($p = 0.25$ against 0.49).

5 Make the rivals compete

Hold real latitude, temperature, rainfall and forest cover constant. Ask again.

3 What we found

Only the country changed

THE FOUR IT DRAWS WARMEST



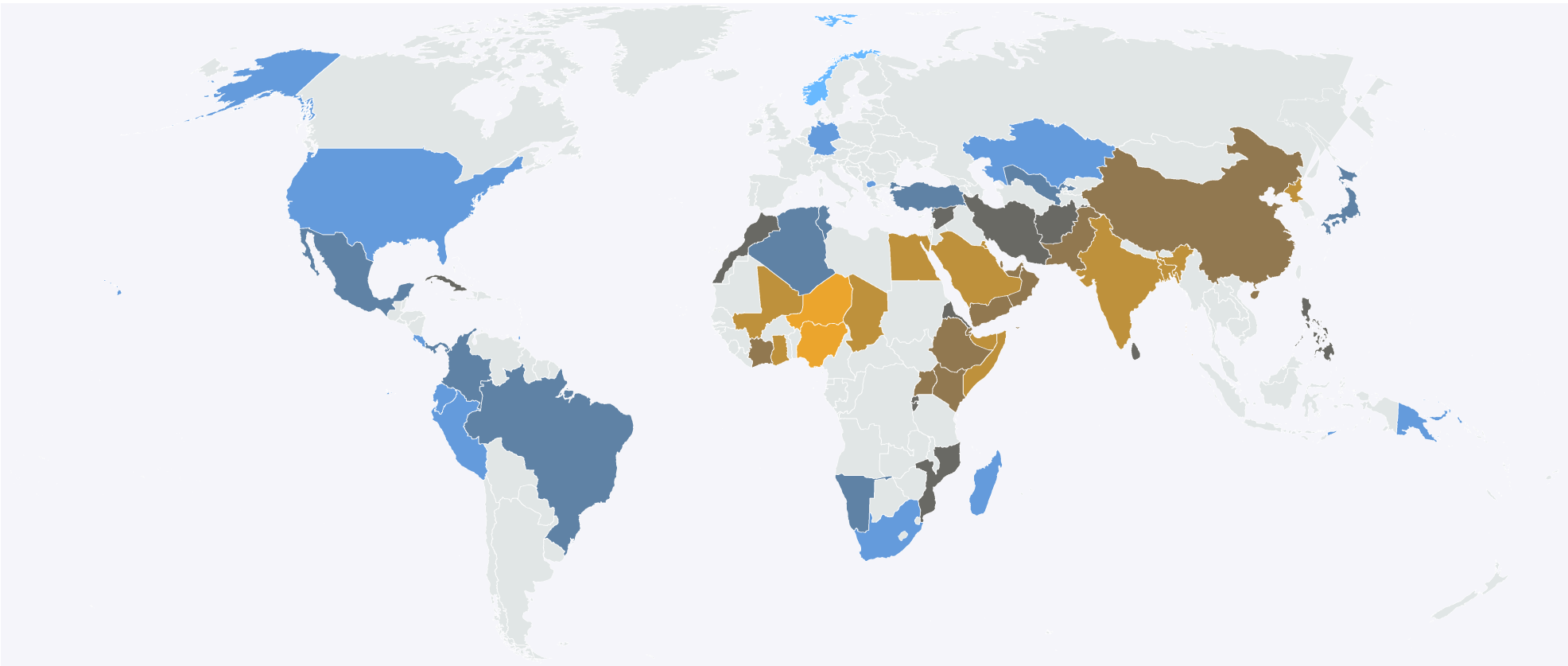
What you are looking at Image 1 of every prompt uses the same seed, so all eight start from identical noise and the street recedes the same way. Only the country named differs. Figures are standard deviations from the average country.

It really does grade countries
53 of 60 beat its own no-country default. Norway alone is cooler.

Income looks real until climate is held constant
 $d = +0.72$ ($p = .018$) falls to $+0.52$, no better than chance.

Asia survives everything
 $+0.73$ ($p = .011$) after every control, in all six kinds of place. Africa's gap vanishes.

Every country, coloured by its warmth



Amber: warmer than average. Blue: cooler. Pale: not in this run.

THE FOUR IT DRAWS COOLEST



But not the poor ones — the dry ones
Saudi Arabia and Kuwait are high-income and near the top.

Latitude was the wrong confound
It accounts for 5%. Forest cover accounts for 30%.

What accounts for the warmth



What each rival explanation accounts for alone. Latitude — the confound the country list was built to defeat — was the wrong one.

4 What it means

The intuitive claim is wrong
It is not warming poor countries but dry, bare ones — substantially what they look like.

The method mattered more than the result
Naively measured, income was there at $p = .018$. It did not survive. An audit without that control would have published it.

“Substantially” is not “entirely”
A regional residual outlives every control. North Korea is humid and half forested, and the warmest thing on the map.

A measurement needs a zero
Ranking countries says who is warmest. Only a no-country baseline says whether anyone is warmed at all.

5 What is next

- Why Asia? The residual now needs a mechanism.
- A better climate measure — four country averages leave 54% unexplained.
- Other models and languages: this is one model's habit, not the industry's.
- Content, not just colour: who is in frame and what they are doing.

What it cannot show

Straight-line controls on four country averages cannot separate a stereotype from a climate, only say whether a pattern needs more than climate to describe it. 18 comparisons per metric: lean on the strongest results only.